

# Project 8: Low Frequency Apparatus

## Technical Design Appendices

### **Case Western Reserve University**

R. Mohiuddin, B. Monreal, Y.-H. Sun

### **Heidelberg University**

F. Spanier

### **University of Illinois, Urbana-Champaign**

C.-Y. Liu

### **Indiana University**

M. Oueslati, W. Pettus, A. Reine

### **Johannes Gutenberg Universität, Mainz**

S. Böser, M. Fertl, A. Lindman, C. Matthé, B. Mucogllava, R. Reimann, F. Thomas, L. Thorne

### **Karlsruhe Institute of Technology, Institute for Astroparticle Physics**

Th. Thümmeler

### **Lawrence Berkeley National Laboratory**

A. W. P. Poon

### **Lawrence Livermore National Laboratory**

K. Kazkaz

### **Massachusetts Institute of Technology**

J. A. Formaggio, M. Li, J. Peña, J. Stachurska, W. Van De Pontseele

### **Pacific Northwest National Laboratory**

J. Gaison, N. S. Oblath, D. Rosa De Jesús, J. R. Tedeschi, B. A. VanDevender

### **Pennsylvania State University**

M. C. Carmona-Benitez, L. de Viveiros, R. Mueller, A. Ziegler

### **University of Pittsburgh**

P. T. Surukuchi

### **University of Texas, Arlington**

B. J. P. Jones

### **University of Washington**

A. Ashtari Esfahani, C. Claessens, P. J. Doe, S. Enomoto, A. Marsteller, E. Novitski, R. G. H. Robertson, G. Rybka

### **Yale University**

K. M. Heeger, J. A. Nikkel, P. L. Slocum, A. B. Telles, T. E. Weiss

---

## Contents

|    |                          |    |
|----|--------------------------|----|
| 1  | Introduction             | 1  |
| 2  | Status Summary           | 2  |
| 3  | Cavity & Ioffe           | 3  |
| 4  | Main Magnet              | 5  |
| 5  | LHe Cryogenics           | 8  |
| 6  | Upper shield cryogenics  | 10 |
| 7  | Lower shield cryogenics  | 12 |
| 8  | Cold Amplifier Feed      | 15 |
| 9  | Pickup Loops             | 16 |
| 10 | Cavity Support           | 17 |
| 11 | Vacuum System            | 18 |
| 12 | Top Service Volume       | 19 |
| 13 | Superconducting Leads    | 20 |
| 14 | Electron Guns            | 21 |
| 15 | Laser Windows            | 22 |
| 16 | Cryostat                 | 23 |
| 17 | Injection Lines          | 24 |
| 18 | Field Compensation Coils | 25 |
| 19 | Support Structure        | 26 |

# **1 Introduction**

## 2 Status Summary

**Table 1:** LFA module status

| <b>Module</b>            | <b>PoC</b> | <b>CAD Status</b> | <b>Documentation Status</b> |
|--------------------------|------------|-------------------|-----------------------------|
| Cavity & Ioffe           | Hamish     | working           | done                        |
| Main magnet              | Hamish     | working           | done                        |
| LHe cryogenics           | Hamish     | working           | done                        |
| Shield cryogenics        | Hamish     |                   |                             |
| Cold amplifier feed      | Wouter     |                   |                             |
| Pickup loops             | Rick, Luiz |                   |                             |
| Cavity support cylinder  | Hamish     |                   |                             |
| Vacuum system            | Hamish     |                   |                             |
| Top service volume       | Hamish     |                   |                             |
| Superconducting leads    | Hamish     |                   |                             |
| Electron guns            | Hamish     |                   |                             |
| Laser windows            | Ben J.     |                   |                             |
| Cryostat                 | Hamish     |                   |                             |
| Injection line           |            |                   |                             |
| Field compensation coils |            |                   |                             |
| Support structure        |            |                   |                             |

### 3 Cavity & Ioffe

The cavity is a smooth Al cylinder with a conical lower end and an upper end connected to a cavity support tube made of thin-wall stainless steel. The smooth interior is a placeholder because the intended design is a mode-filtered cavity in which circumferential grooves are machined into the cavity wall.

The cavity is also the coil form for a superconducting Ioffe coil that provides the atom trapping field. A Halbach array is no longer considered for this. The conductors are precisely located in machined longitudinal grooves on the outer surface, and they run in a complicated pattern over the cone surface. There are 128 conductors equally spaced, looping in a meander around the surface as a single conductor. The inner wall thickness to the bottom of these grooves is 8 mm, recognizing that 6.24 mm is required to resist collapse under atmospheric pressure, which might occur under upset conditions. The conductor diameter is 1.45 mm, and the total wall thickness is therefore 10 mm.

**Table 2:** Dimensions for prototype 560-MHz cryogenic cavity.

| <b>Ioffe Coil</b>                             |      |    |
|-----------------------------------------------|------|----|
| Number of conductors                          | 128  |    |
| Number of loops                               | 64   |    |
| Conductor diameter                            | 1.45 | mm |
| Distance from conductor center to inner wall  | 8    | mm |
| Distance from groove nadir to inner wall      | 7.27 | mm |
| Current                                       | 5000 | A  |
| Field at inner wall                           | 70   | mT |
| Tolerance for conductor azimuthal placement   |      |    |
| Feed pipe ID                                  | 40   | mm |
| Feed pipe flange diameter                     | ?    | mm |
| Cylinder length (including top flange)        | 4900 | mm |
| Total cavity length (including bottom flange) | 5500 | mm |

The total physical length of the cavity is limited by the size of the direct-winding machine at BNL to 5.5 m. The choice of frequency is also limited by the 710-mm-diameter acceptance of that machine. The cone half-angle is now  $35^\circ$ , which produces a stronger effective field than larger angles. See Table 3.

At the upper end the cavity joins to its support tube by a flange currently envisaged to be sealed by a non-magnetic Helicoflex ring. The flange diameter can be minimized by using external C-clamp bolts rather than a drill pattern.

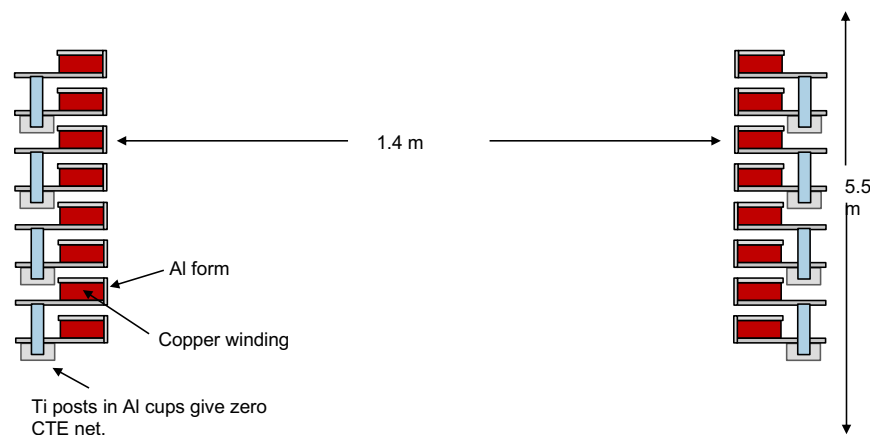
After the Ioffe coil is wound, it is epoxied in place. The cooling is provided by LHe circulating in aluminum tubing loops epoxied to the outer surface of the cavity (or possibly clamped if sufficient conduction can be assured without distortion of tubes or cavity). A single wrap of Al mirror foil or sheet reduces radiant heat absorption from the LN shield.

**Table 3:** Dimensions for prototype 560-MHz cryogenic cavity.

|                                                      |       |     |
|------------------------------------------------------|-------|-----|
| <b>Cavity</b>                                        |       |     |
| Cavity main bore inner radius                        | 326.9 | mm  |
| Cavity wall thickness                                | 10    | mm  |
| Upper flange outer radius                            | 355   | mm  |
| Flange radial width                                  | 28    | mm  |
| Flange thickness                                     | 20    | mm  |
| Cone half-angle                                      | 35    | deg |
| Cone length                                          | 467.0 | mm  |
| Feed pipe extension length (from virtual cone point) | 133.0 | mm  |
| Feed pipe ID                                         | 40    | mm  |
| Feed pipe flange diameter                            | ?     | mm  |
| Cylinder length (including top flange)               | 4900  | mm  |
| Total cavity length (including bottom flange)        | 5500  | mm  |

## 4 Main Magnet

The main magnet is no longer a simple solenoid. Instead it is a stack of 111 identical coils driven by separate power supplies. With suitable currents a field uniform at the  $10^{-7}$  level can be reached even in a ‘short’ solenoid. The structure of such a magnet is indicated in Figure 1.



**Figure 1:** Part of a coil stack comprising the 20-mT background field solenoid (water cooling not shown).

The larger-diameter shaping coils in previous designs are no longer needed. Pinch coils for electron trapping are now effected by increasing the current in selected coils of the main magnet.

Table 4 shows coil and total magnet parameters. The magnet coils are connected in series and energized by 112 independent power supplies. The primary current supply is to be stabilized by NMR or ESR free-induction decay probes positioned between the magnet and the cryostat to achieve the necessary  $10^{-7}$  stability. Each individual coil is set to its exact current with a current-source supply. Powering the coils in this way is convenient and flexible, and leads to the simplest possible (and therefore lowest cost) coil. If the coils have a single winding, the number of turns determines the voltage for a specified number of ampere-turns, which cannot exceed the tolerance of the topmost supply for ground isolation. (An alternative is to include separate secondary windings on each coil to be powered close to ground potential.) The stability of the low-current trim supplies is less stringent, about the  $10^{-5}$  level that is commercially available. The midpoint of the magnet stack should be at the midpoint of the cylindrical part of the cavity, i.e. 2900 mm above the virtual point of the cone.

### Imperfections

**Thermal effects:** A uniform-field solenoid is mostly immune to variations in the radius of coils, whether from mechanical variances or thermal expansion. However, the field is proportional to the linear current density, and therefore the overall length of the solenoid. The spacing of the coils will be made independent of temperature by use of compensated

**Table 4:** Power requirements for main magnet, default conductor Cu. (The power for individual coils near the ends is up to 4 times higher than the midpoint values in the table.)

| Parameter                       | Value (copper)        | Aluminum              | Unit               |
|---------------------------------|-----------------------|-----------------------|--------------------|
| Field                           | 20                    |                       | mT                 |
| Height                          | 5550                  |                       | mm                 |
| Inner radius                    | 700                   |                       | mm                 |
| Coils                           | 111                   |                       |                    |
| Spacing                         | 50                    |                       | mm                 |
| Axial midpoint above cone point | 2900                  |                       | mm                 |
| Current (midpoint)              | 820                   |                       | A-turns            |
| Coil thickness                  | 30                    |                       | mm                 |
| Coil radial span                | 100                   |                       | mm                 |
| Fill factor                     | 0.6                   |                       |                    |
| Effective cross section         | 1800                  |                       | mm <sup>2</sup>    |
| Mean length                     | 4712                  |                       | mm/coil            |
| Maximum total voltage           | 600                   |                       | V                  |
| Resistivity                     | $1.68 \times 10^{-8}$ | $2.65 \times 10^{-8}$ | ohm-m              |
| Resistance per coil             | $4.88 \times 10^{-3}$ | $7.70 \times 10^{-3}$ | ohms               |
| Power                           | 3280                  | 5200                  | W                  |
| Maximum turns per coil          | 149                   | 95                    |                    |
| Density                         | 8850                  | 2700                  | kg m <sup>-3</sup> |
| Mass (conductor only)           | 8333                  | 2542                  | kg                 |

spacers consisting of a titanium rod (Grade 5) and an aluminum cup (T6061). Compensation at 25 C for a 50-mm spacing is achieved with a Ti length of 75.9 mm and an Al cup depth of 25.9 mm. The NMR/ESR stabilizers assure compensation at the next level. The formula for the compensated spacer lengths is

$$L(\text{Ti}) = s \text{ CTE}(\text{Al}) / [\text{CTE}(\text{Al}) - \text{CTE}(\text{Ti})] \quad (1)$$

$$L(\text{Al}) = L(\text{Ti}) - s, \quad (2)$$

where  $s$  is the net spacer length between equivalent points on each succeeding stack. We have used  $\text{CTE}(\text{Al}) = 25.2 \times 10^{-6} \text{ C}^{-1}$  and  $\text{CTE}(\text{Ti}) = 8.6 \times 10^{-6} \text{ C}^{-1}$ .

**Mechanical tolerances:** The most serious imperfection is a curvature in the solenoid, a ‘banana’ distortion, because it introduces an azimuthal asymmetry. The solenoid becomes a section of a torus with the field larger on the small-radius side and conversely. To illustrate quantitatively, the overall solenoid length must be uniform azimuthally to within a wavelength of light. Such precision cannot be achieved by mechanical means, and correction coils consisting of vertical conductors on the exterior of the cryostat or the solenoid will be necessary. The conductors produce a non-uniform transverse field that adds in quadrature with the solenoid field to provide azimuthal correction of the field magnitude. A good mechanical baseline will be accomplished with precision-ground Ti-Al spacers individually measured by optical means and randomized or otherwise selected at installation, and the aggregate az-

**Table 5:** Tolerances on dimensions if magnet uniformity is achieved by mechanical precision alone (provisional).

| Parameter                            | Value | Tolerance    | Unit                |
|--------------------------------------|-------|--------------|---------------------|
| Solenoid length, azimuthal variation | 5550  | $\pm 0.0005$ | mm                  |
| Flatness of individual coils         | -     | $\pm 0.05$   | mm                  |
| Radius of individual coils           | 700   | $\pm 1$      | mm                  |
| Circularity of individual coils      | -     | $\pm 1$      | mm                  |
| Thermal expansion of solenoid        | -     | $\pm 1$      | ppm C <sup>-1</sup> |

imuthal variation is expected to be  $< 10^{-5}$ . Table 5 summarizes mechanical tolerances for a solenoid that meets specified field performance.

**Ground motion:** Microseismic, tidal, aeolian, and human-caused ground motion could excite swaying motion in a firmly anchored solenoid.

It should therefore be mounted on a compliant suspension that provides isolation from external periodic forcing. Water-cooling turbulence and fluctuating supply pressure also must be avoided by using gravity feed from a reservoir, or similar measures.

The suspension can be a ring that supports the magnets and is suspended by a set of rods or cables under tension from another ring above that is supported from the floor. Lateral motions of the floor are then not well transmitted to the magnet. Vertical motions would be transmitted but do not induce the swaying motion that is most troublesome.

## 5 LHe Cryogenics

The cavity operates slightly below 4 K by circulating liquid He through pipes epoxied to the outside. Periodic warmup to about 20 K is required to recycle condensed tritium, which motivated the piping model. Tubing also offers good preemption of He superleaks, avoiding welded assemblies as much as possible. Four loops cover the cylinder and extend over the cone. An annular reservoir holds liquid He under reduced pressure at the top of the cylinder. It has 4 drain ports to a VCR coupling and 4 internal risers extending into the space above the liquid level, also with an external VCR coupling. These 8 ports are connected to the independently assembled cavity with the loops epoxied in place. The loops also end in VCR couplings. The loops are connected to the reservoir by short bellows links with VCR fittings. In operation the reservoir drains into the tubes until the liquid and evolved gas return in the riser. The space above the liquid is continuously pumped via Roots blowers to a pressure below atmospheric and the collected gas is returned to ground level.

The reservoir is an annular stainless steel tank with a 0.1-m radial width and a 0.4-m vertical height (Table 6). When half full it holds 50 L. The cooling tubes hold another 5 L.

**Table 6:** Dimensions for LHe reservoir.

| <b>LHe Reservoir</b>         |      |    |
|------------------------------|------|----|
| Inner radius                 | 360  | mm |
| Outer radius                 | 460  | mm |
| Height                       | 400  | mm |
| Base height above cone point | 5600 | mm |
| Volume                       | 103  | L  |
| Drain ports, VCR             | 4    |    |
| Riser ports, VCR             | 4    |    |
| Riser height                 | 300  | mm |

For magnetic reasons it may be necessary to switch to copper for the tank. Copper is also usable for the cooling tubes. The tank must be able to resist collapse under atmospheric pressure. An estimated boiloff rate of 40 L/d means the gas flow rate to the Roots pumps is about 40,000 L/d, or 0.5 L/s. This can be handled with relatively small bore pipe. The viscosity of He at room temperature is 20  $\mu\text{Pa s}$ , and in a 20-mm bore tube 8 m long the pressure drop is only 40 Pa. However, much larger boiloff rates are encountered during cooldown when the liquid boils as fast as it is supplied. If the supply rate is 100 L/h from a dewar, the boiloff rate is then 30 L/s. The pressure drop in the tube is still only 1200 Pa. Because the boiloff tube leads to room temperature, a chicane at shield temperature must be included to reduce thermal radiation into the LHe reservoir.

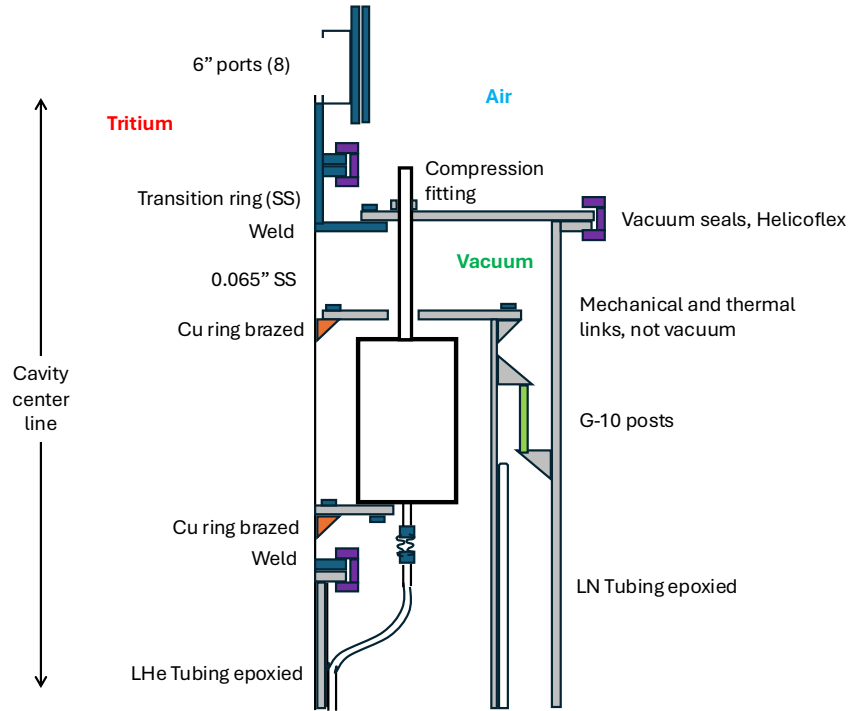
The Roots blowers and reliquefier plant are necessarily at ground level. Liquid He produced at atmospheric pressure and 4.2 K is drawn up a transfer tube to the reservoir about 8 m above the ground and is simultaneously cooled. The transfer tube has a bayonet coupling on the top of the reservoir to deliver liquid above the liquid level in the reservoir. LHe at atmospheric pressure has a density of 125 kg m<sup>-3</sup> and a column 8 m high exerts a pressure

of  $1000 \text{ kg m}^{-2}$ , or 10 kPa. The pressure at the top of the column is then 91 kPa, for which the boiling point of He is 4.05 K. This is not quite low enough to keep the vapor pressure of condensed tritium low enough. To reach 3.5 K, the pressure would be about 50 kPa. A throttle valve is therefore needed on the LHe line, or the supply pressure at the plant needs to be below atmospheric.

A set of radiation shield plates is suspended above the cavity working volume both to reduce external heat input and also to provide the electromagnetic termination of the cavity length. The shield plates need offset holes for pumping, and smaller aligned holes or slots for injection of the calibration electron beam at preselected radial and angular places. The details of these plates are covered in a separate module.

## 6 Upper shield cryogenics

A sketch of the upper shield cryogenics is shown in Fig. 2. For brevity we refer to the shield



**Figure 2:** Sketch of upper shield cryogenics (not to scale).

as the “LN” shield, although it may be preferable to use circulating cold He gas. Dimensions are listed in Table 7. The cavity cap (not shown) is a ring with 8 6-inch ports for bringing in instrumentation to the tritium volume. Since it is part of the tritium system, it is much preferred to make it of aluminum, if suitable ports can be made and sealed.

The order of assembly is as follows:

- Stand up cryostat.
- Install G-10 posts.
- Drop in LN shield with tubes epoxied on.
- Stand up cavity.
- Bolt cavity to thin-wall support cylinder.
- Install split-ring support for LHe reservoir.
- Install reservoir, couple to cavity cooling tubes.
- Install split-ring support for cavity.

**Table 7:** Dimensions upper shield cryogenics.

| <b>Upper shield cryogenics</b>                |       |    |
|-----------------------------------------------|-------|----|
| Cavity main bore inner radius                 | 326.9 | mm |
| Cavity wall thickness                         | 10    | mm |
| Upper flange outer radius                     | 355   | mm |
| Flange radial width                           | 28    | mm |
| Flange thickness                              | 20    | mm |
| LN shield inner radius (Al)                   | 485   | mm |
| LN shield wall                                | 6     | mm |
| LN shield height                              | 6120  | mm |
| Cryostat inner radius (Al)                    | 541   | mm |
| Cryostat wall                                 | 10    | mm |
| Cryostat height                               | 6420  | mm |
| Thin-wall cavity support cylinder (SS) height | 900   | mm |
| Thin-wall thickness                           | 1.5   | mm |
| Transition ring (SS) height                   | 100   | mm |
| Cavity cap (SS or Al) height                  | 500   | mm |

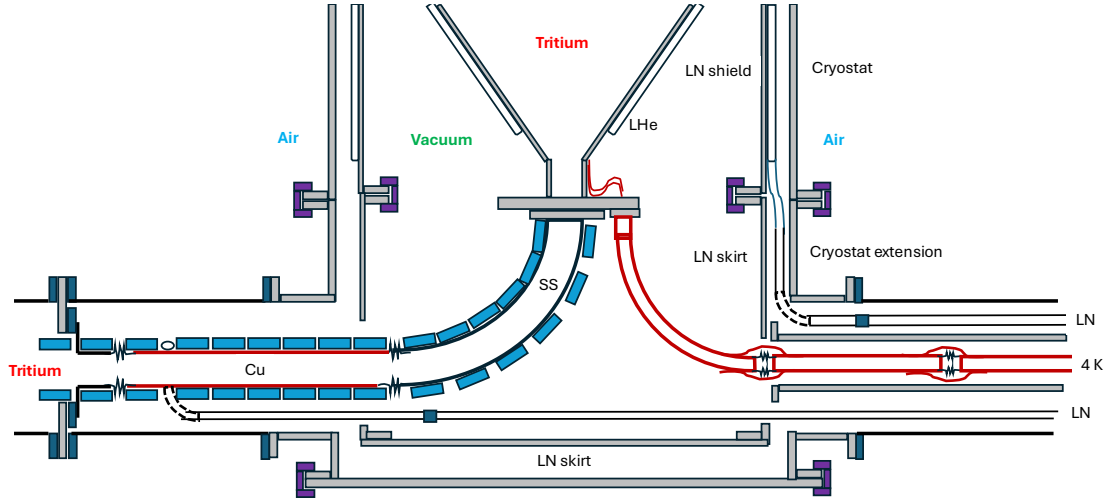
- Drop assembly in to rest on LN shield.
- Drop in annular plate with vacuum seals.
- Attach cavity cap with vacuum seal.

This assumes an assembly area with sufficient vertical crane height is available for assembly ( $\sim 12$  m hook height). Horizontal assembly is possible with suitable fixtures but looks much more difficult.

Both the cryostat and the LN shield are simple Al cylinders, rolled and welded, with flanges. The cryostat welds must be vacuum-tight, but the LN shield is not a vacuum containment. For this reason the LN shield thickness can be less, 6 mm. The inner and outer surfaces of the LN shield should be polished to reduce radiant heat. The outer surface is further protected from radiant heat by a wrap of shiny Al, as is the cavity outer surface, which has epoxied tubes.

## 7 Lower shield cryogenics

A sketch of the lower shield cryogenics is shown in Fig. 3. Many services are brought in at



**Figure 3:** Sketch of lower shield cryogenics (not to scale).

the lower end of the apparatus. The atomic tritium line (when implemented) enters from the evaporative cooling MECB line. The 5-kA superconducting leads for the Ioffe coil, its error-compensation coil, and the axial compensation coils, are brought in from a separate cold box (not shown here) equipped with a pulse-tube cooler and HTS leads for each coil. LN lines for cooling the shield and cooling the tilt section of the magnetic guide are connected here to a supply tank or to a gaseous He cooler.

A major driver of the design is the need to allow for the contraction of the cavity and Ioffe magnet when cooled to operating temperature, about 3.5 K. The contraction is about 24 mm. A straight bellows connection is not satisfactory because the guide magnets and the SC cable would not follow. Instead we introduce tilt sections with bellows joints having only a few ( $\sim 3$ ) convolutions. Very short bellows sections are needed in the magnetic guide because the field drops quickly in the gaps. We lack a full analysis of the curved guide with its gaps. It may be necessary to shape the magnets to reduce the spaces. The magnet mounting is not detailed yet. The magnets should not be attached to the curved beam pipe directly because it must be at 4 K where it meets the cavity and the radiative heating of the magnets from the LN-temperature surroundings will contribute a heat load and a cooling delay after recycling. An independent framework for mounting the magnets is needed. The design of the framework depends on the size and number of the magnets, not yet determined.

We also do not know yet if the SC leads can pass through a tilt section like this. The section must be at 4 K and is cooled both from the cavity end and from the cold box. Braids over the bellows sections cool the tilt section.

Some dimensions are listed in Table 7.

The order of assembly is as follows:

**Table 8:** Dimensions lower shield cryogenics.

| <b>Lower shield cryogenics</b>          |     |    |
|-----------------------------------------|-----|----|
| Inner diameter of curved magnetic guide | 40  | mm |
| Inner diameter of curved copper conduit | 25  | mm |
| Length of tilt sections                 | 800 | mm |
| Diameter of nipple extensions           | 200 | mm |
| Height of cryostat extension            | 500 | mm |

- Stand up cryostat with cavity and Ioffe coil installed.
- Bench assemble the magnetic guide section from its left-hand flange to the horizontal flange.
- Position the cryostat extension below the cryostat. Do not connect the nipples on either side.
- Bring the cryostat extension into position against the cryostat and make the seal.
- Attach the LN skirt, but do not close the bottom plate yet.
- Feed the magnetic guide section through the left-hand port in an arc to clear the openings in the cryostat extension and the LN skirt.
- Join the magnetic guide to the flange of the cavity.
- Bring the LH nipple up and attach it to the cryostat extension. Attach the magnetic guide to the double seal ring. The bolts will be inside the tritium system, which is not ideal – a better scheme is welcome.
- Feed the curved copper conduit through the RH port.
- The SC leads from the magnet will be flying leads, we assume. Pass them through the copper conduit.
- Attach the copper conduit to the Al flange of the cavity. This is not a vacuum connection.
- Bring up the RH nipple and cold box but do not attach yet.
- Connect the LN flexible bellows line VCR to the supply line.
- Connect the RH nipple.
- Connect the LN supply that crosses to the LH side for cooling the tilt section (VCR).
- Close the bottom plate of the LN skirt.
- Close the cryostat extension bottom plate.

Some provision for draining LHe from the Ioffe reservoir will be needed but has not yet been designed. It might be realized as an additional drain line leading from the reservoir to this lower cryogenic enclosure and out the RHS to the cold box where there would be a valve in the line to be opened for draining and a 50-L reservoir that could be pressurized for returning the liquid when recycling was complete.

## 8 Cold Amplifier Feed

## 9 Pickup Loops

## 10 Cavity Support

## 11 Vacuum System

## 12 Top Service Volume

The volume between the top of the cavity and the top plate of the tritium enclosure contains many kinds of instrumentation:

1. The cavity readout is likely to be located here. It may consist of several pickup loops driven by the cavity magnetic component, which in TE011 is radial at the cavity ends. The loops need to be ganged together to a single readout line, and the superconducting transmission line goes out to ground level and the quantum-electronics front end in a dilution refrigerator. A disadvantage of this readout mode is sensitivity to the TE012 mode which is quite close in frequency to TE011. Equatorial readout avoids this but is more difficult to realize in the cavity because of the Ioffe conductors embedded in the wall and the tritium containment requirements.
2. A calibration method uses an electron beam of precisely known energy that Mott-scatters from purposely introduced He gas in the cavity. In the CCA this egun source is movable over cm-scale distances to explore the cavity response. That method appears to be much more challenging in the LFA because of the large distances and the location of the translation mechanism inside a tritium volume. It represents a single-point failure mode, and any failure entails a major intervention. Instead we propose to install a large number of eguns at fixed positions. The translation mechanism is avoided, and a measure of redundancy in case of filament failure, for example, is introduced.
3. A second calibration is an injection line for  $^{83m}\text{Kr}$ . The cavity is too cold to support an equilibrium density but some decays (about  $10^{-6}$ ) will occur as the atoms are in flight before they hit the wall. The injection line must be maintained at no less than 80 K. The same injection line could presumably be used for He injection for the Mott scattering measurements.
4. During the Li operation phase of the LFA, laser diagnostic beams will be used to measure density and temperature. The beams enter from the top, are reflected from mirrors attached to the cone at the bottom of the cavity and re-emerge at the top. Laser windows and clear passage holes in the thermal-radiation-blocking plates that isolate the 3.5-K must be provided.
5. The electron trap has still to be designed. The currents in separate windings of the main magnet can be energized to make a trap, but it may also be advantageous to include one or more coils of smaller diameter in the space above the cavity. These would presumably be superconducting and would require current feedthroughs.

## 13 Superconducting Leads

## 14 Electron Guns

## 15 Laser Windows

## 16 Cryostat

## 17 Injection Lines

## 18 Field Compensation Coils

## 19 Support Structure

### References